2.8 The Quotient Rule

You missed this lesson. Copy this into your notes. That is all you need to do.

Question of the day

How do you differentiate one function divided by another?

Targets

- I can state the quotient rule.
- I can differentiate a quotient and find its slope at a point.
- I can use a table of values with the rule.

Concepts

- **The quotient rule** (h? = (g f? − f g?) / g^2): Bottom times top?, minus top times bottom?. All over the bottom squared.
- **A false rule** (h? ? f? / g?): The derivative of a quotient is not f?/g?. Check it against the tangent slope.
- **Choosing a method** ($(3x^2 + 2?x)/x = 3x + 2x^{-1}?^2$): Simplify first when the bottom is one term. Check with a calculator derivative.

What the class predicted, and what is true

- A student guesses that for $h = \dfrac{\sin x}{x^2}$ the derivative is $\dfrac{\cos x}{2x}$. At $x = 1$ that gives +0.27, but the graph of h is falling there. The guess is... **wrong: the sign disagrees with the graph.** The graph falls at x = 1, so the slope is negative. A positive value cannot be right.
- The curve $y = \dfrac{1}{x}$ for $x > 0$ falls as x increases. Its slope at $x = 2$ is... **negative.** The curve is falling, so the slope is negative.

Huddle notes (answers filled in)

Definitions

- **Quotient rule:** If h = f/g, then h? = (**g f? − f g?**) / **g^2**.
- **Order:** The **bottom** times the derivative of the **top** comes first.
- **Shortcut:** If the bottom is one term, **simplify** first and use the power rule.

Worked example

Find h?(5) for $h(x) = (x^2 + 1)/(x - 3)$.

- g f? − f g? = $(x - 3)(2x) - (x^2 + 1)(1)$
- At x = 5: $2 \cdot 10 - $ **26** = **−6**
- Divide by g^2 = **4**: h?(5) = **−3/2**

Label the rule

1. The numerator is **bottom times top?** minus top times bottom?.
2. Order matters: **subtraction** is not commutative.
3. The denominator is the bottom **squared**.

One worked sample of each skill

Differentiate a quotient

Level 1. Differentiate with the quotient rule.

Answer: $\frac{-5}{(x-2)^2}$ Numerator: $(x-2)(1)-(x+3)(1)=-5$. So $\dfrac{-5}{(x-2)^2}$.

Level 2. Differentiate with the quotient rule.

Answer: $\frac{x^2+2x}{(x+1)^2}$ $\dfrac{(x+1)(2x)-x^2}{(x+1)^2}=\dfrac{x^2+2x}{(x+1)^2}$.

Level 3. Differentiate. Name f and g first.

Answer: $\frac{x^2\cdot e^x-2x^1\cdot e^x}{x^4}$ $\dfrac{x^2e^x-e^x\cdot 2x^1}{x^4}$.

Level 4. Differentiate. You do not have to simplify.

Answer: $\frac{x^5\cos(x)-5x^4\sin(x)}{x^{10}}$ $\dfrac{x^5\cos x-5x^4\sin x}{x^{10}}$.

Slope at a point

Level 1. Let $h(x)=\dfrac{x}{x+2}$. Find $h'(2)$.

Answer: 0.125 $h'(x)=\dfrac{(x+2)-x}{(x+2)^2}=\dfrac{2}{(x+2)^2}$, so $h'(2)=\dfrac{2}{16}=0.125$.

Level 2. Let $h(x)=\dfrac{x^2+5}{x-1}$. Find $h'(3)$.

Answer: -0.5 $h'(3)=\dfrac{(2)(6)-(14)(1)}{4}=-0.5$.

Level 3. A table gives $f(4)=-5$, $f'(4)=2$, $g(4)=10$ and $g'(4)=-6$. Let $h=\dfrac{f}{g}$. Find $h'(4)$.

Answer: -0.1 $h'(4)=\dfrac{(10)(2)-(-5)(-6)}{(10)^2}=-0.1$.

Level 4. A table gives $f(3)=8$, $f'(3)=-5$, $g(3)=10$ and $g'(3)=4$. Let $h=\dfrac{f}{g}$. Find $h'(3)$.

Answer: -0.82 $h'(3)=\dfrac{(10)(-5)-(8)(4)}{(10)^2}=-0.82$.

Simplify first

Level 1. Divide first, then differentiate.

Answer: 4 $\dfrac{4x^2+4x}{x}=4x+4$, whose derivative is 4.

Level 2. Divide first, then differentiate.

Answer: $4x$ $2x^2+7$ has derivative $4x$.

Level 3. Divide first, then differentiate.
Answer: $5 - \frac{2}{2x^{\frac{3}{2}}}$ $5x + 2x^{-1/2}$ has derivative $5 - \frac{2}{2}x^{-3/2}$.

Level 4. Divide first, then differentiate.
Answer: $5 + \frac{2}{x^3}$ $5x + 5 - 1x^{-2}$ has derivative $5 + \dfrac{2}{x^3}$.

Horizontal tangents

Level 1. Let $h(x) = \dfrac{x}{x^2+4}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 2 $h'(x) = \dfrac{(x^2+4) - x(2x)}{(x^2+4)^2} = \dfrac{4-x^2}{(x^2+4)^2}$, which is 0 at $x = 2$.

Level 2. Let $h(x) = \dfrac{x}{x^2+4}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 2 The numerator is $4 - x^2$, which is 0 at $x = 2$.

Level 3. Let $h(x) = \dfrac{x^2+25}{x}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 5 The numerator is $x^2 - 25$, which is 0 at $x = 5$.

Level 4. Let $h(x) = \dfrac{e^x}{x^6}$ for $x > 0$. At which x is the tangent horizontal?
Answer: 6 $h'(x) = \dfrac{e^x x^5(x-6)}{x^{12}}$, which is 0 at $x = 6$ (e? is never 0).